



## **VAST Data Knowledge Base**

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# Overview of VAST DataEngine

VAST DataEngine is the VAST compute orchestration framework. The framework enables users to write, deploy and manage execution pipelines against scheduled or event driven triggers. VAST DataEngine runs on VAST Cluster and dynamically provisions infrastructure and runtime resources according to demand. VAST DataEngine frees users from the need to manage and handle infrastructure scaling and availability or worry about performance.

VAST DataEngine features a web-based graphic user interface, a command line interface and a REST API service for configuring and managing triggers, functions and pipelines and for scheduling, running and monitoring pipelines.

VAST DataEngine features a telemetries service, providing observability over user executed pipelines and functions. The telemetries service collects logs, and traces of pipelines and enables users to query them.

VAST DataEngine can support limitless scope of use cases and a limitless scope of personas who might interact with the framework.

VAST DataEngine utilizes an event broker to stream events into functions. You can choose to use a third party event broker or the VAST Event Broker.

VAST DataEngine is enabled per tenant and can be enabled and configured by a *Cluster Admin* user or a *Tenant Admin* user. A user type called *application user* can be provisioned to non administrative users in order to grant them access to VAST DataEngine.

Deployment of VAST DataEngine involves connecting a VAST Cluster tenant to two external services: a container registry that stores images of functions to be deployed, and a Kubernetes cluster to consume the event broker events and execute the user functions with the above events.

VAST DataEngine supports functions written in Python3. Triggers can be set on any file name suffix and prefix and functions can be written to handle any type of file format. You can also build and deploy HTTP server images.

# Enabling DataEngine on a VAST Cluster Tenant

## Overview

VAST DataEngine enablement requires prerequisite configurations that are external to the VAST cluster as well as prerequisite configurations on the VAST cluster itself.

Additionally, there are configurations on the VAST cluster that are required for DataEngine usage and are not prerequisites to enablement.

## High-Level Steps

To enable VAST DataEngine on a VAST Cluster tenant, follow these steps in this order:

1. [Establish prerequisite external services](#).
2. [Configure Prerequisites on the VAST Cluster](#). These are configurations that you must be established on the VAST Cluster tenant before you enable DataEngine.
3. [Enable DataEngine](#). In this step, you will be prompted to connect the prerequisite external services to the tenant.

## Establish Prerequisite External Services

VAST DataEngine connects with the following services that are external to the VAST Cluster. They must be configured before you start:

- One or more [container registries](#). Used to store images of the functions deployed by DataEngine. The container registry must be accessible over https and have a valid CA certificate.
- One or more [Kubernetes clusters](#). Used to execute DataEngine pipelines and functions.

- Optionally, a third-party event broker. DataEngine requires an event broker. This can be either a third-party event broker or the VAST Event Broker. If you choose to deploy DataEngine with the VAST Event Broker, you do not need to configure it before the enablement procedure.

## Preparing Container Registries

A container registry is a service for storing and distributing container images. You can use a registry service that supports authentication with user name and password, such as a Docker Hub hosted registry service, or a registry that supports a stable authentication secret, such as a AWS ECR hosted registry. You will need to connect the tenant's DataEngine to at least one container registry. You will store images of the functions you want to deploy in the container registry.

## Preparing Kubernetes Clusters

Install the VAST Zarf DataEngine package on each Kubernetes cluster in your environment that you intend to use to deploy VAST DataEngine functions and pipelines. This package installs the following services on the Kubernetes cluster:

- VAST Operator. VAST's Kubernetes operator, which enables deployment of VAST pipelines and functions on the Kubernetes cluster.
- VAST Telemetry Collector. Provides observability logs and traces over pipelines and functions executed on the Kubernetes cluster by DataEngine.
- *Knative* components used by DataEngine.

For environments where Zarf is not already in use for managing Kubernetes deployments, you will also install the Zarf agent and the VAST Zarf mutator from the package.

To install the VAST Zarf DataEngine package on each Kubernetes cluster, do one of the following:

- [Installing the VAST Zarf DataEngine package \(Zarf Not Already in Use\)](#)
- [Installing the VAST Zarf DataEngine package \(Zarf Already in Use\)](#)

## Installing the VAST Zarf DataEngine package (Zarf Not Already in Use)

Use this procedure if Zarf is not already actively being used to manage your Kubernetes cluster deployment.

## Note

To check if Zarf is actively being used to manage your Kubernetes cluster deployment:

| Plaintext                            | <u>Copy</u> |         |          |     |
|--------------------------------------|-------------|---------|----------|-----|
| \$ kubectl get pods -n zarf          |             |         |          |     |
| NAME                                 | READY       | STATUS  | RESTARTS | AGE |
| agent-hook-6c97484f98-5q9jj          | 1/1         | Running | 0        | 10m |
| agent-hook-6c97484f98-tgcjf          | 1/1         | Running | 0        | 10m |
| zarf-docker-registry-fd8859657-4rlm9 | 1/1         | Running | 0        | 11m |

If Zarf is in fact already deployed, switch to [Installing the VAST Zarf DataEngine package \(Zarf Already in Use\)](#).

1. Ensure you have a valid *Kubeconfig* file for connecting to the Kubernetes cluster.
2. Verify that the `inotify` limits are large enough to allow for VAST Zarf DataEngine package installation:

- To check current limits:

| Plaintext                                                                          | Copy |
|------------------------------------------------------------------------------------|------|
| <pre>sysctl fs.inotify.max_user_instances sysctl fs.inotify.max_user_watches</pre> |      |

- To increase the limits temporarily:

| Plaintext                                                                                                | Copy |
|----------------------------------------------------------------------------------------------------------|------|
| <pre>sudo sysctl fs.inotify.max_user_instances=8192 sudo sysctl fs.inotify.max_user_watches=524288</pre> |      |

- To make the change permanent:



| Plaintext                                                                                                                                                                   | Copy |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>echo "fs.inotify.max_user_instances=8192"   sudo tee -a /etc/sysctl.conf echo "fs.inotify.max_user_watches=524288"   sudo tee -a /etc/sysctl.conf sudo sysctl -p</pre> |      |

- Obtain the VAST Zarf DataEngine package, `vast_dataengine_release_<release number>_<pipeline id>.tar.gz`.

#### Note

If you are running a version of VAST Cluster prior to 5.4.1-sp2, run this command before you proceed:

| Plaintext                                                                                                                                          | Copy |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>if [ -f vast-zarf-mutator-amd64-v0.60.0.tar.zst ]; then   mv vast-zarf-mutator-amd64-v0.60.0.tar.zst zarf-init-amd64-v0.60.0.tar.zst fi</pre> |      |

- Extract the Zarf agent init package from the VAST Zarf DataEngine package (for example `zarf-init-amd64-v0.60.0.tar.zst`) and run `zarf init` with the following options :

| Plaintext                                                                                                                                | Copy |
|------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>zarf init --architecture amd64 zarf-init-amd64-v0.60.0.tar.zst - -set REGISTRY_HPA_AUTO_SIZE=true --confirm --log-level debug</pre> |      |

#### Note

The above call assumes that a default storage class exists. Otherwise, add the `--storage-class` option to the call. For example `--storage-class=local-path`.

- Add the following labels to the Kubernetes namespace:

| Plaintext                                                                                                                                                                                                                                                                                         | Copy |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>create_namespace_for_zarf(){   local namespace=\$1   kubectl create ns \$namespace    true   kubectl label namespace \$namespace zarf.dev/vast=mutate } create_namespace_for_zarf vast-dataengine create_namespace_for_zarf knative-eventing create_namespace_for_zarf knative-serving</pre> |      |

6. Extract the VAST Operator file (for example, `zarf-package-dataengine-amd64-1.0.0.tar.zst` ) and run `zarf package deploy` with the following options:

| Plaintext                                                                                                         | Copy |
|-------------------------------------------------------------------------------------------------------------------|------|
| <pre>zarf package deploy --architecture amd64 zarf-package- dataengine*.tar.zst --confirm --log-level debug</pre> |      |

## Installing the VAST Zarf DataEngine package (Zarf Already in Use)

### Note

This procedure can also be used to update the VAST Zarf DataEngine package.

Use this procedure if Zarf is already in use for managing your Kubernetes cluster deployment.

1. Ensure you have a valid *Kubeconfig* file for connecting to the Kubernetes cluster.
2. Verify that the `inotify` limits are large enough to allow for VAST Zarf DataEngine package installation:
  - To check current limits:

| Plaintext                                                                          | Copy |
|------------------------------------------------------------------------------------|------|
| <pre>sysctl fs.inotify.max_user_instances sysctl fs.inotify.max_user_watches</pre> |      |

- To increase the limits temporarily:

| Plaintext                                                                                                | Copy |
|----------------------------------------------------------------------------------------------------------|------|
| <pre>sudo sysctl fs.inotify.max_user_instances=8192 sudo sysctl fs.inotify.max_user_watches=524288</pre> |      |

- To make the change permanent:

| Plaintext                                                                                                                                                                   | Copy |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>echo "fs.inotify.max_user_instances=8192"   sudo tee -a /etc/sysctl.conf echo "fs.inotify.max_user_watches=524288"   sudo tee -a /etc/sysctl.conf sudo sysctl -p</pre> |      |

3. Obtain the VAST Zarf DataEngine package, `vast_dataengine_release_<release number>_<pipeline id>.tar.gz` .
4. Extract the VAST Operator file (for example, `zarf-package-dataengine-amd64-1.0.0.tar.zst` ) and run `zarf package deploy` with the following options:

| Plaintext                                                                                                               | Copy |
|-------------------------------------------------------------------------------------------------------------------------|------|
| <pre>zarf package deploy --architecture amd64 zarf-package- dataengine-amd64*.tar.zst --confirm --log-level debug</pre> |      |

## Configure Prerequisites on the VAST Cluster

This procedure uses the VAST Web UI to create all the prerequisite configurations necessary on a given VAST Cluster tenant for enabling DataEngine.

- If you want to enable DataEngine on a new tenant rather than the default tenant or another existing tenant, [create a new tenant on the \(VAST\) cluster](#).
- If you want to use a third-party event broker as the default event broker for DataEngine, [configure the third-party event broker](#).
- If you want to use VAST Event Broker as the default event broker for DataEngine, [Provision a Bucket Owner User](#)

## Create a New Tenant (Optional)

If you want to enable DataEngine on a new tenant rather than the default tenant or another existing tenant, create a new tenant on the (VAST) cluster.

To create a new tenant with minimal settings:

1. From the left navigation menu, select Element Store and then Tenants.
2. Click Create Tenant to open the Create New Tenant dialog.
3. In the General tab, complete the fields:

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tenant name | Enter a name for the tenant.                                                                                                                                                                                                                                                                                                                                                                                                           |
| Domain      | <p>The domain name for the tenant.<br/>The domain name is used to build the cluster's VMS login URL for the tenant, as shown in the preview below: <code>https://&lt;VMS IP&gt;/#/login/&lt;domain name&gt;</code> .<br/>If a domain name is not specified, the tenant name is used: <code>https://&lt;VMS IP&gt;/#/login/&lt;tenant name&gt;</code> .</p> <div> <p><b>Note</b></p> <p>The domain name is case-insensitive.</p> </div> |

4. Click Create. The tenant is created and appears in the list of tenants in the Tenants page.

## Configure a Third-Party Event Broker

### Note

If you are going to use VAST Event Broker as the default event broker for DataEngine, you can skip this step.

1. Go to Settings -> Notifications -> Notification Kafka Brokers.
2. Click Add a New Kafka Broker and complete these fields:

|        |                                                                                                                                                                                                     |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name   | Enter a name for the event broker configuration.                                                                                                                                                    |
| Tenant | Select one tenant for which the event broker will be available, or leave the default of <i>All tenants</i> so that the broker is available for all tenants.                                         |
| Host   | Enter the bootstrap URL of the event broker server. You can specify an IP or FQDN.<br>If the Kafka cluster runs multiple event brokers, click +Add to add more hosts. You can add up to five hosts. |
| Port   | Enter a port to communicate with the event broker server.                                                                                                                                           |

#### Tip

Ensure that the hosts are accessible from the VAST cluster's management interface at the specified ports.

### 3. Click Add Kafka Broker.

The newly created event broker configuration is added to the list of event brokers.

## Provision a Bucket Owner User

#### Note

If you are going to configure an event broker external to the cluster as the default event broker for DataEngine, you can skip this step.

If you are going to use a VAST Event Broker (a view with the "Kafka" protocol enabled) as the default event broker that DataEngine will use to stream events, you need to provision at least one user with S3 access keys and permission to create buckets on the tenant. This is because when you are enabling DataEngine, you will create the VAST Event Broker and you will need to reference a user as a bucket owner for the view.

You will also need to reference a user of this type as a bucket owner when you [provision S3 bucket views](#) from which triggers can consume events, although this is not an essential step for enabling DataEngine.

In order to be able to grant multiple users to access DataEngine and to create triggers using the same VAST Event Broker, you also need to make sure this user belongs to a group that you will use for all non VMS-manager users that you want to allow to access VAST DataEngine on the tenant, once it's enabled. This is because you will need to give those users bucket listing permission for the VAST Event Broker view (this is done in the next step).

1. Make sure the provider you want to use is attached to the tenant. The provider can be a VAST provider, LDAP, or Active Directory. The provider should be configured on the cluster. For general information about managing providers, see [Providers](#).

#### Note

If you use a VAST provider, enable the provider for *cluster admin* users and *tenant admin* users to manage and view, as needed. The default VAST provider is initially enabled for management by *cluster admin* users only.

To attach a provider to a tenant:

- In the Tenants tab of the Element Store page, right-click the tenant and select Edit.
  - Under Providers and User Access, select the provider that you want to connect to the tenant.
  - Click Update.
2. Create a user on the provider. Grant the user *bucket create* and *bucket delete* permissions.

#### Tip

It is usually best practice that the user belongs to a group designated for granting DataEngine access. You will be able to grant the user's group bucket listing permission for a VAST Event Broker and you will be able to grant the same group [access to DataEngine](#).

#### Tip

To create a user as part of a group on a VAST provider:

1. From the left navigation menu, select User Management and then, under Users and Groups, Local Groups.
2. On the Groups page that opens, click + Create Group.
3. In the Add Group dialog, complete the fields:

| Field                     | Description                                                       |
|---------------------------|-------------------------------------------------------------------|
| Name (required)           | Enter a name for the group.                                       |
| GID (required)            | Enter a POSIX group ID (GID) for the group.                       |
| Local Provider (required) | Select the VAST provider with which the group will be associated. |

- Click Create. The group is created.
- Switch to the Local Users tab.
- Click Create User and complete the following fields:

| Field                             | Description                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name (required)                   | The user name.                                                                                                                                                                                                                        |
| UID                               | The user's POSIX UID.                                                                                                                                                                                                                 |
| Local Provider (required)         | Select the VAST provider with which the user will be associated.                                                                                                                                                                      |
| Leading group                     | The name of the user's leading group.<br>This is the group assigned by default as the owning group of any files created by the user.<br>Select the group you created in the previous step from the dropdown.                          |
| Groups                            | Names of other groups that the user belongs to beside the leading group. Also known as auxiliary groups.<br>Select groups from the dropdown. If a group has not been added to the VAST provider, <a href="#">add the group</a> first. |
| Select tenant to see user details | Select the tenant from the list. Tenants associated with the selected VAST provider (if any) are shown, as well as the default tenant.                                                                                                |
| Allow Create Bucket               | Enable this setting. (Alternatively, you can grant this permission through an identity policy attached to either the group or the user.)                                                                                              |
| Allow Delete Bucket               | Enable this setting. (Alternatively, you can grant this permission through an identity policy attached to either the group or the user.)                                                                                              |

3. Do one of the following, as appropriate, to grant the user an S3 access key pair:
  - Edit the user you just created (if its a local user) and choose to generate an access key pair.
  - Query the user (this is applicable to a user on any provider) and generate an access key pair for the user:
    1. In the Query User or Group tab of the User Management page, query the user on the relevant provider.
    2. In the query result pane on the right, under Access Keys, click Add New Key.

## Enable DataEngine

This procedure enables DataEngine on a tenant. The procedure includes selecting or creating an event broker and event broker topics, connecting the tenant to a container registry and connecting the tenant to a k8s cluster.

### Enabling Data Engine from the VAST Web UI

1. From the left navigation menu, select Element Store and then Tenants.
2. Right-click the tenant and select Enable DataEngine.  
The Enable DataEngine dialog opens.
3. In the Assign Kafka step screen, from the Default Broker dropdown, do one of the following:
  - To use a third-party event broker, select the event broker from the dropdown.
  - To use a VAST Event Broker, select Add new broker.

#### Note

If the VAST Event Broker is already created, you can select it from the dropdown.

The Add View dialog opens. Some of the fields are pre-filled. Continue with the following steps to configure a VAST Event Broker:

1. Complete the following fields:



|                        |                                                                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path                   | Specify the path in the tenant's Element Store where the Event Broker should reside. The path must begin with a slash.                                         |
| Create directory       | Enable this.                                                                                                                                                   |
| S3 bucket name         | Enter a name for the S3 bucket in which the VAST Event Broker will reside.                                                                                     |
| Bucket Owner           | Provide the user you created in <a href="#">Provision a Bucket Owner User</a> . Click to enter the name and start typing and then select a name from the list. |
| Authentication Methods | Set these as you require. For information about these settings, see <a href="#">Configuring User Authentication for the VAST Event Broker View</a> .           |

2. Create or select a suitable view policy. The view policy needs to have S3 Native security flavor enabled. The policy should also grant S3 bucket listing permission to the user group that you are using to provision application users, although you could add the bucket listing permission at a later stage. Bucket listing permission will enable all group members to create triggers using the event broker you are creating.

- To use an existing policy, select the policy from the Policy name dropdown.
- To create a new view policy:
  1. In the Policy name dropdown, select Add new Policy.
  2. In the Add Policy dialog, fill the following fields:

|                         |                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Tenant                  | Select the relevant tenant.                                                                                              |
| Name                    | Enter a name for the policy.                                                                                             |
| Security Flavor         | Select S3 Native.                                                                                                        |
| Group membership source | Select any of the options. This is a required field for view policies, although all options are valid for this use case. |

3. In the S3 section, in the Bucket listing permission (groups) field, enter the name of the user group that you are going to use to provision users' access to DataEngine.

#### Note

This step is not strictly mandatory for enabling DataEngine. It is typically needed, assuming you are using a VAST Event Broker and granting multiple users access to DataEngine. Giving all users bucket listing permission in this policy will enable the users to create triggers using a VAST Event Broker created with a group member as the bucket owner.

#### 4. Click Create.

The policy is created and is selected for you in the Policy name field of the Add View dialog.

3. Create or select a virtual IP pool for the event broker. The virtual IP pool needs to include at least as many virtual IP addresses as there are CNodes on the cluster. It also needs to be dedicated exclusively to the VAST Event Broker.
  - To use an existing virtual IP pool, select the virtual IP pool from the VIP Pool dropdown.
  - To create a new virtual IP pool:
    1. In the Kafka section, from the VIP Pool dropdown, select Add new VIP pool.
    2. In the Add Virtual IP Pool dialog, fill the following fields:

| VAST Web UI Field Name                  | Requirement                                                                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tenant                                  | Specify the tenant on which you are going to enable DataEngine.                                                                                                                                   |
| Name                                    | Give this a suitable name for identifying the virtual IP pool. This virtual IP pool must be used exclusively for the VAST Event Broker.                                                           |
| Role                                    | Set this to Protocols.                                                                                                                                                                            |
| Subnet CIDR<br>IPv4/Subnet CIDR<br>IPv6 | Set this as needed for the IP address range that you supply.                                                                                                                                      |
| IP Range List                           | Supply a range of IP addresses. Include at least as many IPs as there are CNodes on the cluster. This pool of IP addresses will be dedicated exclusively to the VAST Event Broker for the tenant. |

3. Click Create.

The virtual IP pool is created and is selected for you in the VIP Pool field of the Add View dialog.

4. Click Create.

4. From the Default Topic dropdown, select Add new topic to create the default topic on the event broker. Triggers will be streamed to this topic by default and it will be available immediately for hosting triggers when DataEngine is enabled. In most cases, there is no need to create additional topics for hosting triggers. Assuming the default topic will be used for all or most triggers, it's advisable to set wide partitioning, probably in the region of 50-100 partitions.

5. In the Create Topic dialog, complete the following required fields:

|            |                             |
|------------|-----------------------------|
| Topic Name | Enter a name for the topic. |
|------------|-----------------------------|

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of partitions | <p>Enter the number of partitions for the topic.</p> <p>Each partition can hold up to 1000 topics. The number of partitions in a topic cannot be changed after the topic has been created.</p> <p>The default topic will be available immediately for hosting triggers when DataEngine is enabled. Unless your use case specifically requires multiple topics, the default topic may be used for hosting all triggers. Therefore, it is usually advisable to set wide partitioning, probably in the region of 50-100 partitions.</p> |
| Retention period     | <p>Specify the amount of time to keep an event record in the topic. When the retention period for a record expires, the record is deleted from the topic.</p> <p>The default retention period is seven days. The minimum allowed retention period is 6 hours.</p>                                                                                                                                                                                                                                                                    |
| Compaction           | <p>If this flag is set, VAST Cluster keeps only the latest record version for each key in the partition log. Any previous versions of the record are deleted.</p> <div> <p><b>Note</b></p> <p>Compaction is performed asynchronously in the background. At some point in time, duplicate keys may exist.</p> </div> <p>Select this checkbox to enable compaction in the topic. With compaction, the topic preserves only the latest version of each message key in each partition.</p>                                               |
| Advanced Options     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Timestamp source     | <p>Select the source to use for the event time stamp:</p> <ul style="list-style-type: none"> <li>Time event is created (producer). Sets the event timestamp based on the time when the event was encountered at the event producer.</li> <li>Time event is logged (server). Sets the event timestamp based on the time when the event record was added to the log at the event broker.</li> </ul>                                                                                                                                    |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event time evaluation period | <p>The maximum acceptable time between the event time and server time.</p> <ul style="list-style-type: none"> <li>• Infinity. For Time event is created (producer) timestamps, this option specifies that any difference between the producer time and the server time is acceptable, e.g. the message timestamp can be earlier or later than the broker timestamp for any amount of time.</li> <li>• Define Manually. For Time event is created (producer) timestamps, specifies the acceptable difference between the producer time and the server time: <ul style="list-style-type: none"> <li>◦ Not before: Determines how much earlier the message timestamp can be than the broker timestamp. If this value is exceeded, the message is rejected.</li> <li>◦ Not after: Determines how much later the message timestamp can be than the broker timestamp. If this value is exceeded, the message is rejected.</li> </ul> </li> </ul> |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

6. Click Create.
7. In the Default Deadletter Topic dropdown, select Add new topic to create the deadletter topic, which is the topic to which events are routed in case of function failures, routing failures and dispatching failures. The deadletter topic will not be available to configure as the topic for any trigger event.
8. Click Add Certificate to upload a TLS certificate.
9. Paste a CA certificate into the box provided and click Upload Certificate.
10. Click Next.
11. In the Link Kubernetes Cluster step, complete these fields:

|                           |                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------|
| Name (required)           | Enter a name for the Kubernetes cluster to which you want DataEngine to connect.                 |
| API Server URL (required) | Enter the Kubernetes API server endpoint URL.<br>For example: <i>https://kube-server-57:6443</i> |
| Description               | Enter a description of the Kubernetes cluster. This is optional.                                 |

12. In the Certificates area, use the fields provided to upload the CA certificate, client certificate and private key used for authenticating over mTLS to the Kubernetes cluster. For each certificate, click +Add new , paste the certificate content, [base64-encoded](#), and then click Save Client certificate /Save CA certificate/Save Private Key.

13. In the Tags area, enter any metadata tag that you want to tag the Kubernetes cluster resource with, and select Add Tag to add the tag. Repeat for additional tags. This is optional.
14. Click Next.
15. In the Select Namespaces step, the namespaces on the Kubernetes cluster are listed on the left under All possible Properties. Use the selection boxes and arrow buttons to choose which of the namespaces to allow the tenant's DataEngine to access and list them on the right under Selected Properties. Pipelines will be deployed in the selected namespace(s).
16. Click Next.
17. In the Link Container Registry step, complete the fields to connect the container registry:

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary kubernetes cluster     | Select the primary Kubernetes cluster from the dropdown.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Additional kubernetes clusters | Select any additional Kubernetes clusters that are configured on the tenant if you want them to be able to deploy any function that has its image stored on the container registry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Name                           | Enter a name for the container registry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Base URL                       | Enter the base URL of the container registry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Description                    | Enter a description for the container registry. (Optional.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Authentication Method          | <p>Select the authentication method required by the container registry for the DataEngine to authenticate to the container registry:</p> <ul style="list-style-type: none"> <li>• User credentials. Choose this option if a user name and password is required. In the Username and Password fields, enter valid user credentials for authenticating to the container registry.</li> <li>• Kubernetes secret. Choose this option if a stable authentication secret is required. In the Kubernetes Secret Name field, enter the name of an existing Kubernetes Secret that contains the credentials for authenticating to the container registry .</li> <li>• None. Choose this option if no authentication is required to connect to the container registry.</li> </ul> |
| Tags                           | <p>Optionally, you can use this section to tag the container registry with any meaningful text strings for ease of organization and identification.</p> <p>To add a tag, enter a string in the Tag field and then click Add Tag. The tag is added to the Tags list.</p> <p>If you want to remove a tag that you added, click the tag's removal icon.</p>                                                                                                                                                                                                                                                                                                                                                                                                                |

18. Click Finish.

# Provisioning User Access and Permissions for DataEngine

## Overview

### User Types

You can provision the following types of users to enable access to DataEngine on a given tenant:

- *Tenant admin* users. As administrators with access to a specific tenant's VMS for management purposes, these users can be authorized to enable DataEngine on the tenant, add Kubernetes clusters and container registries, and create buckets to trigger events. They automatically have permission to use all DataEngine features.
- *Application user*. This is a type of user that you can provision for users in your organization who have no administration role on the cluster or the tenant. This type of user can access DataEngine on the tenant without any of the VMS tenant administration functionality. These users do not have automatic permission to use all DataEngine features. Specific permissions are granted through identity policies. These include creating and managing triggers, serverless functions, and pipelines, deploying pipelines, and observing telemetry logs and traces.

### Provisioning Application Users

Application users are provisioned through user groups on authentication providers. The following provider types are supported:

- VAST providers.



#### Note

To enable *cluster admin* and/or *tenant admin* users to configure user provisioning using a local provider, make sure the provider has management enablement for the relevant user type. See {TBD}.

The default VAST provider is initially, by default, enabled for management by *cluster admin* users and not *tenant admin* users.

- LDAP
- Active Directory

## Granting DataEngine Access and Permissions to Application Users

In order to grant application users access to DataEngine, you add the users to a group. The group must be on a provider that is connected to the tenant. In the tenant configuration, the group must be explicitly granted Data Engine access.

In order to perform tasks such as creating triggers and pipelines on DataEngine, *application users* must be granted permissions through identity policies.

The procedure below guides you to select an option within the tenant configuration to add the group to a predefined DataEngine role and to assign the group to a predefined identity policy that grants specific permissions.

## Provisioning Tenant Admin Users for DataEngine

*Tenant admin* type users automatically have access to Data Engine regardless of their specific VMS permissions.

For the ability to create and manage source views (buckets) for triggering events, a *Tenant admin* user requires permission for the *Logical* realm.

For the ability to set up DataEngine provisioning on the tenant, a *Tenant admin* user requires permission for the *Security* realm.

For details about VMS administrator realms and how to create a *Tenant admin* user, see [VMS Administrator Users and Permissions](#).

# Provisioning Application Users for DataEngine

This procedure creates a user group on a provider that has access DataEngine, adds users to the group and grants permissions to the users in the group to perform DataEngine tasks such as creating triggers and functions and building pipelines.

1. Create a user group on a provider.

### Tip

If you are using a local (VAST) provider, you can do the following to create a user group:

1. From the left navigation menu, choose User Management and then Local Groups.
2. Click Create Local Group and complete the following fields:

| Field                     | Description                                                       |
|---------------------------|-------------------------------------------------------------------|
| Name (required)           | The group name.                                                   |
| GID                       | A GID number for the user.                                        |
| Local Provider (required) | Select the VAST provider with which the group will be associated. |

3. Click Create. The group is created.

2. Connect the provider to the tenant and add the user group to the tenant for Data Engine access:
  1. In the Tenants tab of the Element Store page, right-click the tenant and select Edit.
  2. Under Providers and Users Access, select the provider on which you created the group.
  3. Under Who Can Access This Tenant (Data Engine), in the Users Group field, start typing the name of the user group and select the group from the autocomplete suggestions.
  4. Enable Assign Group to DataEngine role. This setting adds the group to a VMS role that is created automatically for DataEngine.

#### Note

This setting is technically optional. If you do not select this option, you will need to separately add the user group to a VMS RBAC role as described in [Managing Administrative Roles in VAST Web UI](#). This is because membership of a role that belongs to the tenant is required to enable login access to the tenant.

5. Enable Assign DataEngine identity policy to group. This setting assigns the data-engine- *TENANT\_NAME* identity policy to the group, where *TENANT\_NAME* is the name of the tenant. This policy grants users permission to create and manage DataEngine resources.

#### Note

Enabling this policy is optional. If you choose not to enable it, the group will not have any permissions to perform any Data Engine related tasks until you assign another identity policy to the group. See [Configuring Identity Policies for DataEngine](#).

The content of the data-engine- *TENANT\_NAME* identity policy is:

| Plaintext                                                                                                                                                                                                                                                                                                                                                 | Copy |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>{   "Id": "DataEnginePolicy1757876601",   "Version": "2012-10-17",   "Statement": [     {       "Sid": "DataengineTablesAccess",       "Action": [         "s3:HeadBucket",         "s3:Tabular*Transaction",         "s3:TabularList*",         "s3:TabularGet*",         "s3:TabularQueryData"       ],       "Effect": "Allow",     }   ] }</pre> |      |

```

    "Resource": [
      "dataengine-*",
      "dataengine-*/*"
    ]
  },
  {
    "Sid": "DataEngineDefault",
    "Action": [
      "dataengine:CreateTrigger",
      "dataengine:CreateFunction",
      "dataengine:CreatePipeline"
    ],
    "Effect": "Allow",
    "Resource": [
      "vast:dataengine:triggers:*",
      "vast:dataengine:functions:*",
      "vast:dataengine:pipelines:*"
    ]
  }
]
}

```

#### Note

The `CreateTrigger`, `CreateFunction` and `CreatePipeline` actions allow the user to create triggers, functions and pipelines and also to update, modify and view those triggers, functions and pipelines that the user created. The `CreatePipeline` action includes permission to configure function deployment.

6. Click Update.
3. Create users on the provider. Assign them to the user group. If the user is on a local provider, assign a password.

#### Tip

If you are using a local provider, you can do this to create a user:

1. From the left navigation menu, choose User Management and then Users.
2. In the Users page that opens, click Create User and complete the following fields:

| Field                             | Description                                                                                                                                                                                                                            |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name (required)                   | The user name.                                                                                                                                                                                                                         |
| UID                               | The user's POSIX UID.                                                                                                                                                                                                                  |
| Local Provider (required)         | Select the local provider with which the user will be associated.                                                                                                                                                                      |
| Leading group                     | The name of the user's leading group.<br>This is the group assigned by default as the owning group of any files created by the user.<br>Select the group you created in the previous step from the dropdown.                           |
| Groups                            | Names of other groups that the user belongs to beside the leading group. Also known as auxiliary groups.<br>Select groups from the dropdown. If a group has not been added to the local provider, <a href="#">add the group</a> first. |
| Select tenant to see user details | Select the tenant from the list. Tenants associated with the selected local provider (if any) are shown, as well as the default tenant.                                                                                                |
| Allow Create Bucket               | Enable this setting.                                                                                                                                                                                                                   |
| Allow Delete Bucket               | Enable this setting.                                                                                                                                                                                                                   |

3. In the User Password section, enter a temporary password for the user, or click Generate Password to have one randomly generated. The password can be any string matching the password requirements set in the VMS Settings. The user will be required to change the password when they first log in using it. If the user has access to multiple tenants, the same password is used to login to all of them.
4. Click Create. The user is created.

4. Grant each user an S3 access key pair. See [Managing S3 User Access from the VAST Web UI](#).

#### Note

S3 access key pairs are per tenant, even when a user belongs to multiple tenants that all use the same provider.

Application users can also manage their own S3 access key pairs. If the user does not have s3 access keys, the user will be prompted to create a pair after logging into the tenant.

5. Attach identity policies as needed to the user and/or user group. If you applied the data-engine-*TENANT\_NAME* identity policy to the user group you specified for DataEngine access in the tenant configuration, you may not need to apply further identity policies. See [Configuring Identity Policies for DataEngine](#).

## Configuring Identity Policies for DataEngine

DataEngine requires permissions to create buckets and to perform queries against VAST databases. For application users, these are granted through identity policies.

The predefined data-engine-*TENANT\_NAME* identity policy grants users all permissions needed for DataEngine tasks. In the tenant configuration, if you select the Assign DataEngine identity policy to group option, this policy is applied to all the users in the user group specified under Who Can Access This Tenant (Data Engine) (as described in step 2 of the procedure described above).

If for some reason, you don't choose to apply that identity policy to the user group, you can create custom identity policies and attach them to each user or to the group. For general information about creating and modifying identity policies and assigning them to users and groups, see [Creating Identity Policies](#).

#### Note

The *vast-data-engine-collector-policy* is automatically created and applied to the DataEngine telemetry collector, to enable the collector to query DataEngine telemetries.

## DataEngine Permissions Granted through Identity Policies

The following supported S3 actions apply to DataEngine resources (triggers, functions and pipelines):

| Action                                    | Description      |
|-------------------------------------------|------------------|
| <code>dataengine:CreateTrigger</code>     | Create trigger   |
| <code>dataengine&gt;DeleteTrigger</code>  | Delete trigger   |
| <code>dataengine:GetTrigger</code>        | List triggers    |
| <code>dataengine:UpdateTrigger</code>     | Modify triggers  |
| <code>dataengine:CreateFunction</code>    | Create function  |
| <code>dataengine&gt;DeleteFunction</code> | Delete function  |
| <code>dataengine:GetFunction</code>       | List Functions   |
| <code>dataengine:UpdateFunction</code>    | Modify Function  |
| <code>dataengine:CreatePipeline</code>    | Create pipelines |
| <code>dataengine&gt;DeletePipeline</code> | Delete pipelines |
| <code>dataengine:GetPipeline</code>       | List pipelines   |
| <code>dataengine:UpdatePipeline</code>    | Modify pipelines |

# Provisioning Source Views for Trigger Events

DataEngine triggers may be *element* type triggers, which watch for events related to elements in a *source view*. This includes creation and deletion of S3 objects or S3 object tags. In order that DataEngine users can create element triggers, a source view must be created to host those trigger events and it must be accessible to DataEngine users.

Application users cannot create source views. This must be done in VMS by a manager user (a *tenant admin* or *cluster admin* user).

To provide a source view, [create a view](#) with the following configuration:

| VAST Web UI Field Name | Requirement                                                                                                                                                                                                                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path                   | Specify the path in the tenant's Element Store.                                                                                                                                                                                                                                                                  |
| Create directory       | Enable this.                                                                                                                                                                                                                                                                                                     |
| Policy name            | Select a view policy on the same tenant with <i>S3 Native</i> security flavor.<br><br>To ensure that all <i>application users</i> can access the bucket, add the same user group that you are using to provision application users to the <i>S3 bucket listing permission</i> field in the policy configuration. |
| Protocols              | Set this to S3 Bucket.                                                                                                                                                                                                                                                                                           |
| S3 bucket name         | Enter a name for the bucket.                                                                                                                                                                                                                                                                                     |
| Bucket Owner           | Provide the user you created in <a href="#">Enabling Data Engine on a VAST Cluster Tenant</a> . Click to enter the name and start typing and then select a name from the list.                                                                                                                                   |





You can create and otherwise manage these views as described in [Managing Views](#).

# Managing Kubernetes Clusters

During the initial enablement of Data Engine, you are prompted to configure one Kubernetes cluster. After enablement, you can modify the configuration of the default cluster that you created and you can configure additional Kubernetes clusters.

## Viewing Kubernetes Clusters

1. From the left navigation menu, select Data Engine and then Kubernetes Clusters.
2. The following fields are displayed for each Kubernetes cluster:

|            |                                                                |
|------------|----------------------------------------------------------------|
| Name       | The name of the Kubernetes cluster resource.                   |
| API Server | The API server endpoint URL of the Kubernetes cluster.         |
| GUID       | A globally unique VAST ID for the Kubernetes cluster resource. |
| Tenant     | The tenant to which the Kubernetes cluster resource belongs.   |
| Tags       | Any metadata tags added to the Kubernetes cluster resource.    |

## Adding a Kubernetes Cluster

1. From the left navigation menu, select Data Engine and then Kubernetes Clusters.
2. Click Link a New Cluster or right-click anywhere in the list area and select Create New.
3. Complete the following fields:

|                           |                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------|
| Name (required)           | Enter a name for the Kubernetes cluster to which you want DataEngine to connect.                 |
| API Server URL (required) | Enter the Kubernetes API server endpoint URL.<br>For example: <i>https://kube-server-57:6443</i> |
| Description               | Enter a description of the Kubernetes cluster. This is optional.                                 |

4. In the Certificates area, use the fields provided to upload the CA certificate, client certificate and private key used for authenticating over mTLS to the Kubernetes cluster. For each certificate, click +Add new , paste the certificate content, base64 encoded, and then click Save Client certificate /Save CA certificate/Save Private Key.
5. In the Tags area, enter any metadata tag that you want to tag the Kubernetes cluster resource with, and select Add Tag to add the tag. Repeat for additional tags. This is optional.
6. Click Next.
7. In the Select Namespaces screen, the namespaces on the Kubernetes cluster are listed on the left under All possible Properties. Use the selection boxes and arrow buttons to choose which of the namespaces to allow the tenant's DataEngine to access and list them on the right under Selected Properties. Pipelines will be deployed in the selected namespace(s).
8. Click Done.

## Modifying Kubernetes Clusters

To modify a Kubernetes cluster configuration, right-click the Kubernetes cluster and select Edit.

## Deleting Kubernetes Clusters

To delete a Kubernetes cluster, right-click the Kubernetes cluster and select Remove.

# Managing Container Registries

During the enablement of Data Engine, one container registry is configured. You can add each additional container registry afterwards.

## Viewing Container Registries

1. From the left navigation menu, select Data Engine and then Container Registries.
2. The following fields are displayed for each container registry:

|                    |                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Name               | The name of the container registry resource.                                                                                              |
| Base URL           | The base URL of the container registry.                                                                                                   |
| GUID               | A globally unique VAST ID for the container registry resource.                                                                            |
| Kubernetes Cluster | The Kubernetes cluster linked to the container registry, on which functions that reference images on the container registry are deployed. |
| Tenant             | The tenant to which the container registry resource belongs.                                                                              |
| Tags               | Any metadata tags added to the container registry resource.                                                                               |

## Adding a Container Registry

1. From the left navigation menu, select Data Engine and then Container Registries.
2. Right-click in the list area and select Create New.
3. Complete the fields to connect the container registry:

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary kubernetes cluster     | Select the primary Kubernetes cluster from the dropdown.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Additional kubernetes clusters | Select any additional Kubernetes clusters that are configured on the tenant if you want them to be able to deploy any function that has its image stored on the container registry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Name                           | Enter a name for the container registry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Base URL                       | Enter the base URL of the container registry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Description                    | Enter a description for the container registry. (Optional.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Authentication Method          | <p>Select the authentication method required by the container registry for the DataEngine to authenticate to the container registry:</p> <ul style="list-style-type: none"> <li>• User credentials. Choose this option if a user name and password is required. In the Username and Password fields, enter valid user credentials for authenticating to the container registry.</li> <li>• Kubernetes secret. Choose this option if a stable authentication secret is required. In the Kubernetes Secret Name field, enter the name of an existing Kubernetes Secret that contains the credentials for authenticating to the container registry .</li> <li>• None. Choose this option if no authentication is required to connect to the container registry.</li> </ul> |
| Tags                           | <p>Optionally, you can use this section to tag the container registry with any meaningful text strings for ease of organization and identification.</p> <p>To add a tag, enter a string in the Tag field and then click Add Tag.</p> <p>The tag is added to the Tags list.</p> <p>If you want to remove a tag that you added, click the tag's removal icon.</p>                                                                                                                                                                                                                                                                                                                                                                                                         |

4. Click Done.

## Modifying a Container Registry

To modify a container registry configuration, right-click the container registry and select Edit.

## Deleting a Container Registry

To delete a container registry, right-click the container registry and select Remove.

## Disabling DataEngine

To disable DataEngine:

1. From the left navigation menu, select Element Store and then Tenants.
2. Right-click the tenant and select DataEngine and then Disable.
3. In the confirmation box, click Disable again to confirm the action.

# Glossary of Terms

## VAST AI OS Terms

The VAST DataEngine is built on top of the VAST AI OS. The following terms are features of the VAST AI OS platform that occur in this guide.

### Element

A logical data construct for things the scale of a file or object. The VAST DataStore currently supports the following types of Elements: files, objects, tables, event broker topics and vectors, block volumes. In the current offering of VAST DataEngine, all elements are S3 objects.

VAST DataEngine interfaces may also refer to DataEngine *triggers* and *functions* as *pipeline elements*.

### Tenant

A resource belonging to a VAST Cluster that isolates a division of cluster storage resources, usually to serve data paths to a client base. DataEngine runs on a VAST Cluster tenant. Your user account accesses VAST DataEngine on a single VAST Cluster tenant.

### VAST Cluster

An instantiation of the VAST AI Operating System.

### VAST DataStore

The universal storage foundation for VAST AI OS. includes object, file, table, and block storage functionality in a single namespace.

### VAST Management Service (VMS)

The management service that runs on a VAST Cluster.

### View

VAST's protocol-independent name for the logical protocol endpoints through which data is shared. A *view* can be an NFS export, an SMB share, an S3 bucket, a block subsystem and/or an other type of endpoint, depending on the supported access protocols.

## Other Terms Used in This Guide



### Cloud events

Events sent in a format defined by the Cloud Events (CE) open standard specification that defines a common, interoperable way to describe event data, simplifying its delivery across services and platforms.

## About this Guide

This guide describes how to use VAST DataEngine to build and deploy data pipelines on a VAST Cluster tenant.

This guide assumes that VAST DataEngine has been enabled on a tenant of a VAST Cluster. For information about how to enable VAST DataEngine, see the [VAST Cluster 5.4 Administrator's Guide](#) or the [VAST Cluster 5.4 Tenant Administrator's Guide](#).

The instructions in this guide use the VAST DataEngine web user interface wherever available. VAST DataEngine also supports the same functionality through the DataEngine CLI and through the VAST Management Service (VMS) API.

# Overview of VAST DataEngine

VAST DataEngine is a serverless development framework that enables you to manage and deploy functions running against scheduled or event driven triggers, and to build functions and triggers into pipelines, schedule and deploy those pipelines. VAST DataEngine runs on VAST Cluster and dynamically provisions infrastructure and runtime resources according to demand. VAST DataEngine frees users from the need to manage and handle infrastructure scaling and availability or worry about performance.

VAST DataEngine features a web-based graphic user interface, a command line interface and a REST API service for configuring and managing triggers, functions and pipelines and for scheduling, running and monitoring pipelines.

VAST DataEngine features a telemetries service, providing observability over user executed pipelines and functions. The telemetries service collects logs, metrics and traces of pipelines and supports queries.

VAST DataEngine utilizes an event broker to stream triggered events into functions.

VAST DataEngine connects to external container registries, where images of your functions must be stored in order to be deployed, and to external Kubernetes clusters, which consume the event broker events and execute the user functions with the above events.

VAST DataEngine supports functions written in Python3. Triggers can be set on any file name suffix and prefix and functions can be written to handle any type of file format.

## VAST DataEngine Resource Types

VAST DataEngine features the following types of resources:

- **Trigger.** This is a resource that watches for a defined type of event and sends structured events, formatted as *cloud events*, to an event broker topic. VAST DataEngine enables the creation of two types of triggers:

- Element triggers. These triggers watch for events where elements are added or removed from a view or where tags are added or removed from elements in a view.
  - Schedule triggers. These triggers issue events on a schedule.
- Function. A function is a pointer to an image of a function stored on a container registry. VAST provides a CLI command to package a function that you can push to a container registry, ready to be referenced by a DataEngine function.
- Function deployment. A function deployment is an instantiation of a function as part of a pipeline. Each deployment of any function within any pipeline can be configured independently.
- Pipeline. A runtime instantiation of a set of functions and triggers that can be interconnected in a defined flow or in a group of defined flows.

# Access and Permissions

DataEngine is provisioned on a tenant of a VAST Cluster.

To access DataEngine on any given VAST Cluster tenant, you will need a valid user account, provisioned by a VAST Management Service (VMS) cluster or tenant administrator.

Your administrator must grant permissions to your user to enable you to perform actions such as creating, modifying and deleting triggers, functions, function deployments and pipelines.

You will need an S3 access key pair for DataEngine to use to access S3 buckets for various purposes. If you don't have one, you will be prompted to create one when you log into DataEngine. You can manage your own S3 access keys.

# Logging into VAST DataEngine

1. Obtain the following from the administrator of your VAST Cluster or VAST Cluster tenant:
  - The URL for a VAST Cluster tenant.
  - The user name and password of a user with access to DataEngine on the tenant. The password may be temporary.
2. Browse to the URL and log in with your user name and password.

## Note

The first time you log in, you may be prompted to set a new secure password and to create s3 keys:

1. Enter a secure password of your choice in the New password field.  
The password you enter is validated for several security criteria, which are displayed. Ensure that and enter the same password in the Repeat password field. Obtain the login URL for the VAST Cluster tenant to which you have been granted access.
2. Click Save Password or, if you don't have an S3 access key pair, Save and Generate Access Key.  
The password is saved.  
If an access key pair is created for you, copy the secret key to the clipboard and keep a record of it for future use, since the secret key is not retrievable.
3. Click Done.

The VAST DataEngine Dashboard appears.

# Managing Your Access Keys

S3 access keys are required for DataEngine to be able to access source views for element type triggers, to access VAST Event Broker, which is hosted on a *view*, and to browse resources such as functions and triggers.

To manage your access key pairs:

- At the top right of the page, click your Avatar and select Manage Access Keys. Your active S3 access keys are displayed. You can delete keys and generate new keys. When done, click Done.

## To generate a new key

1. Click Generate Another Key.  
The secret key is displayed.
2. Copy the secret key to the clipboard (a button is provided for copying) and store it for safe keeping. You will not be able to retrieve the secret key again.

## To delete a key

Click the trash can icon next to the key.

# Installing the VAST DataEngine CLI

The VAST DataEngine CLI provides commands that perform the same tasks as the VAST DataEngine web user interface. The CLI also enables you to package a function as an image to upload to the container registry. You will need to initialize and package functions in order to deploy them within DataEngine pipelines (as described in [Creating a DataEngine Function](#)).

1. [Download](#) the VAST DataEngine CLI binary file for your operating system.
2. Save the file locally.
3. Run the `vastde config init` command with the following syntax to initialize and edit the CLI configuration:

| Plaintext                                                                                                                                                                                                                  | Copy |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>vastde config init --password PASSWORD                   --tenant TENANT_NAME                   --username USERNAME                   --builder-image-url BUILDER-IMAGE-URL                   --vms-url VMS-URL</pre> |      |

Substitute the arguments for the following values:



|                               |                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PASSWORD</b>               | The password is associated with the username you provide as USERNAME.                                                                                                                                                                                                                                                                                                                        |
| <b>TENANT_NAME</b>            | The name of the tenant you want to connect to (running DataEngine).                                                                                                                                                                                                                                                                                                                          |
| <b>USERNAME</b>               | A username of a user with access to the specified tenant. This can be an <i>application user</i> , a <i>tenant admin</i> user or a <i>cluster admin</i> user.                                                                                                                                                                                                                                |
| <b>BUILDER-<br/>IMAGE-URL</b> | <p>The URL of the VAST DataEngine Builder Docker image, in the format <code>vastdataorg/vast-builder/TAG</code> , where TAG is a version tag.</p> <p>To find a version tag, browse to <code>https://hub.docker.com/r/vastdataorg/vast-builder/</code> and select a tag from the Recent tags dropdown.</p> <p>For example, <code>vastdataorg/vast-builder:v1.0.60ee8cc7.vast-5.4</code> .</p> |
| <b>VMS-URL</b>                | The URL to the cluster's VAST Management Service (VMS) in the format <code>https://VMS_IP</code> , where <code>VMS_IP</code> is the VMS IP.                                                                                                                                                                                                                                                  |

#### 4. To view the content of the configuration file:

|                                 |      |
|---------------------------------|------|
| Plaintext                       | Copy |
| <code>vastde config view</code> |      |

Now that the configuration is initialized, you can run CLI commands against the specified tenant's DataEngine.

#### Note

VAST DataEngine CLI documentation is available at [here](#).

# Creating a Trigger

DataEngine functions are triggered by events. Create triggers to define which events to watch for. A trigger can either produce events on a schedule or watch for events related to objects in a bucket. You can build triggers into a pipeline and so that they trigger functions.

1. From the left navigation menu, select Manage Elements () and then Triggers.
2. Click Create New Trigger.
3. Enter a name for the trigger in the Trigger Name field.
4. In the Trigger Type dropdown, select the type of trigger you want to create:
  - Element. A trigger that watches for events related to elements in a source view.
  - Schedule. A trigger that produces events on a schedule.
5. Complete the relevant fields:
  - If you selected *Element* as the *Trigger Type*:

|             |                                                                                                                                                                                    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | A description for the trigger.                                                                                                                                                     |
| Source view | From the dropdown, select the view in which the events will occur. The dropdown lists all S3 bucket views on the local VAST Cluster tenant to which you have permission to access. |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event type               | <p>From the dropdown, select a type of event.</p> <ul style="list-style-type: none"> <li>◦ ElementCreated. The creation of an element in the source view.</li> <li>◦ ElementDeleted. The deletion of an element from the source view.</li> <li>◦ ElementTagCreated. The creation of a tag on an element in the source view.</li> <li>◦ ElementTagDeleted. The deletion of a tag on an element in the source view.</li> </ul> <div> <b>Note</b><br/> You will be able to restrict the relevant elements by object key prefix and suffix. </div> |
| Source type              | <p>The type of elements that register events for the trigger. This is restricted to S3 objects in the current version. Therefore, the field is read only and displays S3.</p>                                                                                                                                                                                                                                                                                                                                                                  |
| Destination              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Target Event Broker View | <p>From the dropdown, select the view that hosts the VAST Event Broker to use to stream events for the trigger.</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Topic                    | <p>From the dropdown, select the event broker topic that should be used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Filters                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Object Key Filters       | <p>To filter elements by suffix or prefix, enter a prefix and/or a suffix. The event type definition will be restricted to objects with the specified prefix and/or suffix in their object keys.</p>                                                                                                                                                                                                                                                                                                                                           |
| Custom Extensions        | <p>Use this field to add key-value pairs to be passed as event output extensions. Each key can contain only lowercase letters and digits, must start with a letter, and must be 1-20 characters long.</p>                                                                                                                                                                                                                                                                                                                                      |
| Tags                     | <p>Use this field to tag the trigger.</p> <p>To add a tag, click Add Tags, enter a key-value pair in the fields provided and then click Add to Tags.</p>                                                                                                                                                                                                                                                                                                                                                                                       |

- If you selected *Schedule* as the *Trigger Type*:

|                   |                                                                                                                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description       | A description for the trigger.                                                                                                                                                                                                                         |
| Destination       |                                                                                                                                                                                                                                                        |
| Target Kafka View | From the dropdown, select the view that hosts the VAST Event Broker to use to stream events for the trigger.                                                                                                                                           |
| Topic             | From the dropdown, select the event broker topic that should be used.                                                                                                                                                                                  |
| Schedule          | Do either of the following to define a schedule for the trigger: <ul style="list-style-type: none"> <li>◦ Select Simple and use the fields provided to enter a frequency.</li> <li>◦ Select Advanced to type the schedule in Quartz syntax.</li> </ul> |
| Custom Extensions | Use this field to add key-value pairs to be passed as event output extensions. Each key can contain only lowercase letters and digits, must start with a letter, and must be 1-20 characters long.                                                     |
| Tags              | Optionally add tags to the trigger. To add a tag, click Add Tags, enter a key-value pair in the fields provided and then click Add to Tags.                                                                                                            |

## 6. Click Create Trigger.

The trigger is created and is listed in the Triggers page.

# Creating a DataEngine Function

## Overview

Any function that you want to be able to call and deploy within a DataEngine pipeline must first be packaged as an image and stored on one of the container registries that is connected to the DataEngine instance that you are working with.

VAST provides a capability, through the DataEngine CLI, to package your function code as an image ready to store on a container registry. You then push that image to your container registry. You need to obtain the image tag from the container registry for the specific image.

For each image stored on the container registry, you can create a *function* resource within DataEngine. A function, in DataEngine terms, is a resource that points to an image of a function stored on a container registry. A function can be built into a DataEngine *pipeline* and deployed as part of that pipeline. Within a pipeline, a function can be invoked by a trigger or by another function. Each function deployment within a pipeline has a configuration that determines the execution environment provided to the function in that instance.

When you want to update the code for a given function, you can update the code, store the updated image on the container registry and create a new revision of the same function with a different image tag. When you deploy a function in a pipeline, you can choose to deploy a specific revision or the current (latest) revision. When you edit a pipeline, you can switch a function to a different revision.

## Workflow

1. [Prepare the Function Image in the Container Registry](#)
2. [Create the Function Resource](#)

## Prepare the Function Image in the Container Registry

1. Install the [VAST DataEngine CLI](#).
2. Run the `vastde functions init` command to create a scaffold for your function.  
For example:

| Plaintext                                                                     | Copy |
|-------------------------------------------------------------------------------|------|
| <pre>vastde functions init python-pip my_first_function -t ~/functions/</pre> |      |

The output is a directory structure need for implementing your function.  
For example:

| Plaintext                                                                                                                                                                                                                                                                           | Copy |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>~/functions/my-first_function/ ├─ Aptfile          # OS dependencies ├─ README.md        # Project documentation ├─ customDeps       # custom python libraries ├─ main.py          # Handler functions └─ requirements.txt # Python dependencies  0 directories, 5 files</pre> |      |

The files in the directory structure are created empty.

3. Write your code in the files provided to implement the function. See the *VAST DataEngine Runtime SDK Guide*.

The following is sample `main.py` content:

| Plaintext                                                                                                                            | Copy |
|--------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>from sharedlib.module import lib  def init(ctx):     lib.shared_func()  def handler(ctx, event):     return "Hello World"</pre> |      |

4. Run the `vastde functions build` command to package the function as an image and to tag the image. For example:

| Plaintext                                                                                                                                           | Copy |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>FUNC_NAME="function" FUNC_PATH="example/path/to/function" vastde functions build \$FUNC_NAME -target \$FUNC_PATH --image-tag my-function</pre> |      |

5. Test the function locally:
  1. Run the now built function as a local container: `vastde functions localrun`
  2. Send a cloud event to the locally running function for testing and invocation: `vastde functions invoke`
  3. Iterate on code and rebuild as needed.
6. Tag the local image with a tag of your choice:

| Plaintext                                                                         | Copy |
|-----------------------------------------------------------------------------------|------|
| <pre>docker tag \${FUNC_NAME}:latest CONTAINER_REGISTRY/ARTIFACT_SOURCE:TAG</pre> |      |

Substitute the following:

- `CONTAINER_REGISTRY` . The URL of a container registry to which you have privileges to push images. The container registry must also have a connection configured on the tenant for DataEngine. This connection is configured on the VAST Cluster tenant by an administrator.
- `ARTIFACT_SOURCE` . A path on the container registry where you intend to push the image.
- `TAG` . The text that you want to tag the image with (such as the name of the function).

7. Push the image to the container registry:

| Plaintext                                                               | Copy |
|-------------------------------------------------------------------------|------|
| <pre>docker push CONTAINER_REGISTRY/ARTIFACT_SOURCE:\${FUNC_NAME}</pre> |      |

## Create the Function Resource

1. From the left navigation menu, select Manage Elements () and then Functions.
2. Click Create New Function.
3. Complete the fields:

|                      |                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Function Name        | Enter a name for the function.                                                                                                                    |
| Description          | Enter a description for the function.                                                                                                             |
| Revision Alias       | Enter an alias for the initial revision of the function.                                                                                          |
| Revision No.         | This field is a display field. It displays the revision number, which is initially 1. The number increments on every edit of the function.        |
| Revision Description | Enter a description for the revision.                                                                                                             |
| Container Registry   | From the dropdown, select the container registry where the image of the function is stored.                                                       |
| Artifact Source      | Enter the path to the container image.                                                                                                            |
| Image Tag            | Enter the image tag attached to the relevant version of the container image.                                                                      |
| Full image path      | This is the full path to the image. It is formed automatically from the container registry, artifact source and image tag values that you supply. |

4. Click Create Function.  
The function is created and is listed in the Functions page.



# Building and Deploying a Pipeline on VAST DataEngine

## Overview

A pipeline may feature any number of sequences connecting triggers and functions. A sequence typically begins with a trigger, which invokes a function. A function can invoke another function. That function may invoke another function. And so on. Your pipeline can feature multiple sequences. Each function deployment within the pipeline has an independent deployment configuration.

Building a pipeline involves configuring triggers and functions, connecting triggers and functions in sequences, configuring the deployment of each function and finally, deploying the function.

## Workflow

1. Review the types of triggers that you can use:
  - Element triggers. These triggers watch for events where elements are added or removed from a view or where tags are added or removed from elements in a view.
  - Schedule triggers. These triggers issue events on a schedule.
2. Design your pipeline, which will consist of deployment sequences of triggers and functions.
3. Optionally [create triggers](#) and [create functions](#) before you create the pipeline. You can also do this after creating the pipeline resource or, if applicable, use existing triggers and functions.
4. [Create the Pipeline Resource](#)
5. [Build the Pipeline](#). At this stage, you will be able to select triggers and functions that are already available, or create new ones. This stage includes configuring the deployment of each function. You will be able to save a draft of the pipeline at any stage, to save your work if it is not ready to be deployed.
6. [Deploy the Pipeline](#). Once it is deployed, you will be able to observe [logs](#) and [traces](#) through the VAST DataEngine web user interface. You can also leverage the DataEngine API for telemetry observation.

# Create the Pipeline Resource

1. From the left navigation menu, select Pipeline Management.
2. Click Create New Pipeline
3. Complete the fields:

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pipeline Name        | Enter a name for the pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Description          | Enter a description for the pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Kubernetes Cluster   | From the dropdown, select the Kubernetes cluster that you want to use to execute the pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Namespace            | Select which namespace to use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Pipeline Secret Keys | <p>Use this field to pass a secret to all function deployments in the pipeline. The secret can contain multiple key-value pairs, such as access keys for accessing data and metadata of S3 buckets.</p> <div> <p><b>Note</b></p> <p>You can also specify secrets for individual function deployments in their individual configurations.</p> </div> <p>You can enable functions to access these secret keys through use of the VAST DataEngine runtime SDK <code>ctx.secrets</code> function in your function code. You can either enter the secrets manually or you can upload them as a yaml file.</p> <p>To enter secrets manually:</p> <ol style="list-style-type: none"> <li>1. Click Create secret.</li> <li>2. Enter the secret name in the Secret Name field.</li> <li>3. In the Key and Value fields provided, enter one of the keys of the secret and a valid value for it.</li> <li>4. Click the add button provided to add another key-value pair row as needed until you have added all keys needed for the secret.</li> <li>5. To add another secret, click Create Another secret and repeat the above steps.</li> </ol> <p>To upload a secret in a .yaml file, click Import Yaml File.</p> <p>To remove all secrets, click Clear All Secrets.</p> |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environment Variables | <p>Environment variables are key–value pairs that you can define for your functions to access at runtime. They are part of the execution environment provided to your functions.</p> <p>You will be able to define environment variables per function deployment when you build the pipeline.</p> <p>In this field, define any environment variables that you want to provide to <i>all</i> functions in the pipeline sequence. Function level environment variables overwrite pipeline level ones if the key of a key value pair matches and the value does not.</p> <p>To add environment variables, do one of the following:</p> <ul style="list-style-type: none"> <li>• Enter the variables: <ol style="list-style-type: none"> <li>1. Click Add Variable.</li> <li>2. Enter a key in the Key field and a value in the Value field.</li> <li>3. If you want to add another variable, click  and enter another pair.</li> <li>4. Repeat until you have added all the variables you want to add.</li> </ol> </li> <li>• Import environment variables from a file: Click Import ConfigMap.</li> </ul> |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 4. Click Create Pipeline.

The pipeline resource is created and the Visual Builder opens, enabling you to [build the pipeline](#).

## Build the Pipeline

The Visual Builder enables you to add the triggers and functions that you want to build into the pipeline and to connect them to each other. The pipeline can consist of multiple pipeline flows, with triggers invoking functions and functions invoking functions.

### Note

You cannot connect a trigger to another trigger. You cannot connect elements in a loop.

During the process, you can click Save Draft to save a draft at any time. When you are done, you can [deploy the pipeline](#).

The following actions are available to help you build the pipeline:

- [Drag Pipeline Elements into the Builder](#)
- [Search Triggers and Functions](#)
- [Create New Triggers and Functions](#)
- [Connect Triggers and Functions](#)
- [Configure Function Deployment](#)

## Drag Pipeline Elements into the Builder

Drag and drop triggers and functions into the visual builder to build your pipeline:

- To the left of the Visual Builder, select Triggers or Functions to see the library of existing triggers and functions.
- To add a trigger or function to the pipeline, drag it into the Visual Builder from the list on the left.

## Inspect a Trigger or Function in the Visual Builder

To see the properties of any pipeline element in the visual builder, select the element and click the Inspect button to the right of the builder. The element's properties appear on the right.

For a function, the following information is shown: the container registry where the image is stored, the image source, and the image tag.

For a trigger, the following information is shown: the source view, broker, topic, and the type of trigger.

## Search Triggers and Functions

To the left of the Visual Builder, select Triggers or Functions, click the search box and enter a string to search for triggers or functions by name.

## Create New Triggers and Functions

- To create a new trigger, select the Triggers tab at the left of the Visual Builder and click Create New Trigger.  
Follow the procedure [described here](#).

The new trigger is now visible in the library of triggers in the left panel.

- To create a new function, select the Functions tab at the left of the Visual Builder and click Create New Function.

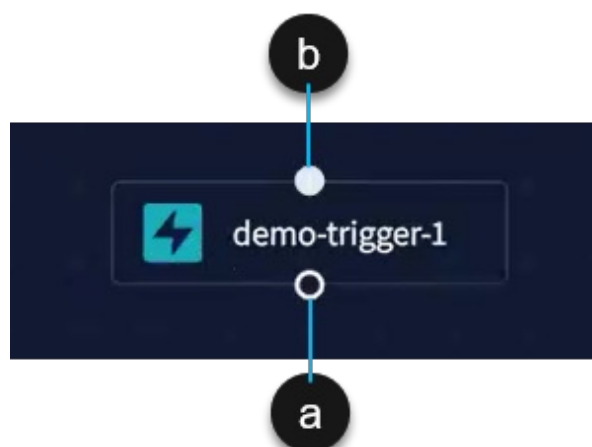
Follow the procedure [here](#).

## Connect Triggers and Functions

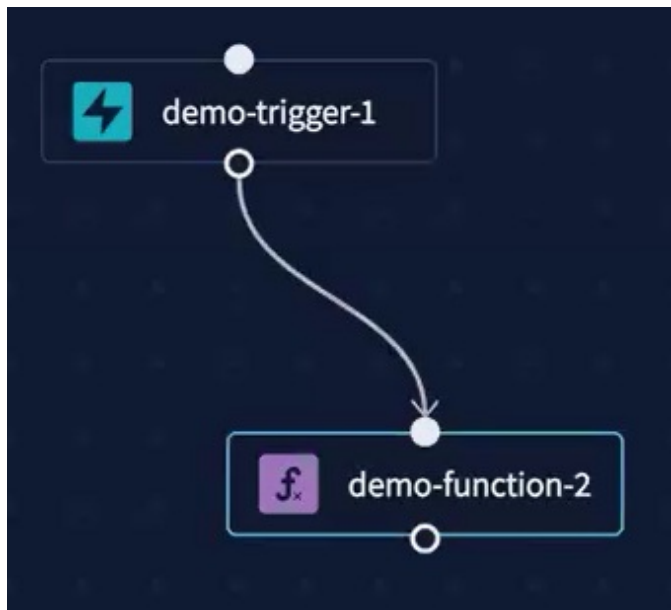
You can build into the pipeline a flow in which a trigger invokes a function. When deployed, the pipeline will execute the function on each event of the trigger. You can also connect a function to another function so that the pipeline executes the additional function on the output of the first function. Within a pipeline, you might have multiple flows, each starting with a different trigger.

### Connecting Elements to Each Other

To connect a trigger or function to a function, drag the open handle (a) of the first trigger or function to the solid handle (like b) of the function.



A connector appears, showing the direction of invocation (the selected trigger will invoke the selected function):



## Disconnecting Elements from Each Other

To disconnect an element from another element, select the arrow and either drag it into the open space or press Backspace or Delete on your keyboard. The link disappears.

## Removing Elements from the Builder

To remove a trigger or function from the builder, select the trigger or function and press the BACKSPACE key on your keyboard.

## Configure Function Deployment

Each function deployment included in the pipeline has its own configuration that determines the execution environment provided to the function. There is a default configuration for every function deployment. You can edit it for each function deployment as needed.

To edit the configuration a function deployment:

1. Select (click) the function in the Visual Builder. The function deployment configuration appears on the right.
2. If you want to change which revision of the function to deploy, select an alternate revision from the Revision Number dropdown.
3. To edit other parameters, click the Edit button next to the section heading, and edit the values in the fields.

You can configure the following parameters:

- [Secret Keys](#)
- [Deployment Resources](#)
- [Deployment Configuration](#)
- [Environment variables](#)

## Secret Keys

Add secret keys to pass a secret to the specific function deployment. The secret can contain multiple key-value pairs, such as access keys for accessing data and metadata of S3 buckets.

### Note

Secret keys that are specified in the pipeline configuration are passed to all function deployments.

You can enable functions to access these secret keys through use of the VAST DataEngine runtime SDK `ctx.secrets` function in your function code. See *The VAST DataEngine Runtime SDK Guide* for details.

To add secrets:

1. Click Create secret.
2. Enter the secret name in the Secret Name field.
3. In the Key and Value fields provided, enter one of the keys of the secret and a valid value for it.
4. Click the add button provided to add another key-value pair row as needed until you have added all keys needed for the secret.
5. To add another secret, click Create Another secret and repeat the above steps.

## Deployment Resources

Most of these resources have minimum and maximum values. Minimum values are guaranteed resource to provide to the function deployment. Maximum values are limits on the maximum resources provided to the function deployment.

|                        |                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Concurrency            | The minimum and maximum number of functions to be deployed on the Kubernetes cluster (the number of pods).                        |
| CPU                    | The minimum CPU to guarantee for the function and the maximum CPU to allow to the function.                                       |
| Memory                 | The minimum memory to guarantee for the function and the maximum memory to allow to the function.                                 |
| Autoscaling RPS factor | The rate of requests per second at which autoscaling should begin.                                                                |
| Disk (Ephemeral)       | This is not provided by default. If the function needs ephemeral storage, enter an amount of capacity to provide to the function. |

## Deployment Configuration

|                    |                                                                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timeout            | The timeout for the function.                                                                                                                                                                                                                       |
| Retries            | The number of retries if the function deployment fails or times out.                                                                                                                                                                                |
| Log level          | <p>The log level in case of failure. Possible values:</p> <ul style="list-style-type: none"> <li>• NOTSET</li> <li>• DEBUG</li> <li>• INFO (default)</li> <li>• WARNING</li> <li>• ERROR</li> <li>• CRITICAL</li> </ul>                             |
| Method of Delivery | <p>Method of processing trigger events. Possible values:</p> <ul style="list-style-type: none"> <li>• ordered. Processes events in the same sequence that they are produced.</li> <li>• unordered. Events may be processed concurrently.</li> </ul> |

## Environment variables



Environment variables are key–value pairs that you can define, for your function to access at runtime. They are part of the execution environment provided to your function.

## Deploy the Pipeline

When you are done building a pipeline with the Visual Builder, click Deploy.

To deploy a pipeline that you saved as a draft, find the draft in the Pipelines tab of the Pipelines Management page. The status shows as *Draft*. Right-click the pipeline and select Deploy.

You can view the status of the pipeline in the Pipelines tab of the Pipelines Management page. Initially, while the pipeline is being deployed, the status is *In Progress*. When deployed, the status changes to Running.

# Managing Triggers

## Viewing Triggers

1. From the left navigation menu, select Manage Elements () and then Triggers.

The following properties are displayed for each trigger:

|             |                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name        | The name of the trigger.                                                                                                                                                                                                     |
| GUID        | A globally unique ID for the trigger.                                                                                                                                                                                        |
| Type        | The type of trigger: <ul style="list-style-type: none"> <li>• Element. A trigger that watches for events related to elements in a source view.</li> <li>• Schedule. A trigger that produces events on a schedule.</li> </ul> |
| Status      | The status of the trigger.                                                                                                                                                                                                   |
| Source View | For an element trigger, this is the view (bucket) that the trigger watches for events.                                                                                                                                       |
| Created At  | The date and time on which the trigger was created.                                                                                                                                                                          |
| Tags        | Metadata tags that were added to the trigger.                                                                                                                                                                                |

2. To see more properties of a given trigger, select the trigger and click Inspect.

## Modifying a Trigger

1. From the left navigation menu, select Manage Elements () and then Triggers.
2. Right-click the trigger and select Edit.
3. Make changes and click Update Trigger.

## Deleting a Trigger

1. From the left navigation menu, select Manage Elements () and then Triggers.

2. Right-click the trigger and select Delete and then confirm the deletion.

# Managing DataEngine Functions

## Viewing Functions

1. From the left navigation menu, select Manage Elements () and then Functions.

The following properties are displayed for each function:

|                  |                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name             | The name of the function. If an expansion arrow appears, click the arrow to see multiple revisions of the same function.                                 |
| GUID             | A globally unique ID for the function.                                                                                                                   |
| Owner            | The owning user of the function                                                                                                                          |
| Created          | The date and time on which the function was created.                                                                                                     |
| Modified         | The date and time on which the function was last modified. When you modify a function, a new revision is created.                                        |
| Current Version  | The latest revision number created for the function. Where there is more than one revision, expand the function in the Name column to list each version. |
| No. Of Revisions | The number of revisions of the function that have been created.                                                                                          |
| Tags             | Metadata tags that were added to the function.                                                                                                           |

2. To see more properties of a given function, select the trigger and click Inspect.

## Modifying a Function

Modifying a function creates a new revision with a new revision number. When you want to update your function code, you can build a new image of the function, tag it as you wish, push it to the container registry and then create a new revision of the function pointing to the updated image.

Creating a new revision does not automatically update the revision used in a pipeline. When you create a new revision to use in a pipeline, you also need to update the pipeline to use the new revision.

There is an option to create a local revision of a function. Local revisions cannot be used in a pipeline.

1. From the left navigation menu, select Manage Elements () and then Functions.
2. Right-click the function and select Edit.

This function's properties appear. The function name and description are not modifiable. The revision number is incremented automatically from the previous revision.

3. Make changes as needed:

|                      |                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision Alias       | An alias for the revision.                                                                                                                        |
| Make local revision  | Check the box to make the revision local. A local revision that cannot be used in a pipeline.                                                     |
| Revision Description | A description specifically for the revision.                                                                                                      |
| Container Registry   | From the dropdown, select the container registry where the image of the function is stored.                                                       |
| Artifact Source      | Enter the path on the container registry where the image resides.                                                                                 |
| Image Tag            | Enter the image tag attached to the relevant version of the container image.                                                                      |
| Full image path      | This is the full path to the image. It is formed automatically from the container registry, artifact source and image tag values that you supply. |

4. Click Update Function.

The new revision is created and listed in the Functions page as a revision of the original function.

## Deleting a Function

1. From the left navigation menu, select Manage Elements () and then Functions.
2. Right-click the function and select Delete and then confirm the deletion.

# Managing Pipelines

## Viewing Pipelines

1. From the left navigation menu, select Manage Pipelines () and then Pipelines.

The following properties are listed for each pipeline:

|            |                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name       | The name of the pipeline resource.                                                                                                                                                                                                                                                                                                                              |
| GUID       | A globally unique ID for the pipeline resource.                                                                                                                                                                                                                                                                                                                 |
| Status     | <p>The status of the pipeline:</p> <ul style="list-style-type: none"> <li>• Failure. Hover over the status to see the reason why the pipeline failed.</li> <li>• Running. The pipeline is deployed and running.</li> <li>• Draft. The pipeline was saved as a draft and is not yet deployed.</li> <li>• In progress. The pipeline is being deployed.</li> </ul> |
| Created At | The time the pipeline resource was created.                                                                                                                                                                                                                                                                                                                     |
| Tags       | Any metadata tags that were added to the pipeline.                                                                                                                                                                                                                                                                                                              |

2. To see more properties of a given pipeline, select the pipeline and click Inspect.

## Modifying a Pipeline's Configuration

1. From the left navigation menu, select Manage Pipelines () and then Pipelines.
2. Right-click the pipeline and select Edit Pipeline.
3. Make changes to the configuration as needed. (You cannot modify the name of the pipeline.)
4. Click Update Pipeline.

## Modifying a Pipeline Deployment

1. From the left navigation menu, select Manage Pipelines () and then Pipelines.
2. Right-click the pipeline and select Go to Builder.
3. Make changes as you like to the way the pipeline is built, such as triggers, functions, sequencing of triggers and functions, and function deployments.
4. When you are done making changes, click Deploy.  
The pipeline is deployed again.

## Deleting a Pipeline

1. From the left navigation menu, select Manage Pipelines () and then Pipelines.
2. Right-click the pipeline and select Delete Pipeline and then confirm the deletion.

# Dashboard Monitoring

The Dashboard displays:

- Numbers: Total triggers, Total functions, and Total pipelines. These numbers reflect all DataEngine resources on the tenant for which you have read permission.
- System-wide Overview: Functions Performance and Trends. Two graphs are displayed:
  - Total Events and Failures. This shows total events and failures in recent minutes. For example, if a deployed pipeline calls a schedule trigger that is scheduled to generate an event every five minutes, those events would appear on the graph at five minute intervals.
  - Average Function Invocation Duration. This shows the average duration aggregated across all running functions.



## Viewing Pipeline Logs

Logs are collected using the Open Telemetry framework.

To view logs:

- To see all logs, open the Pipeline Management page from the left navigation menu and select Logs.

To see logs for a specific pipeline, right-click the pipeline and select Pipeline Logs.

This opens the Logs page and filters the logs for the pipeline you selected.

The Logs page displays the following properties for each log:

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timestamp | The date and time of the logged event.                                                                                                                                                                                                                                                                                                                                                                                                         |
| Level     | The severity level: <ul style="list-style-type: none"> <li>• TRACE. A fine-grained debugging event.</li> <li>• DEBUG. A debugging event.</li> <li>• INFO. An informational event. Indicates that an event happened.</li> <li>• WARN. A warning event. Not an error but is likely more important than an informational event.</li> <li>• ERROR. An error event.</li> <li>• FATAL. A fatal error such as application or system crash.</li> </ul> |
| Scope     | The scope that emitted the log: <ul style="list-style-type: none"> <li>• user</li> <li>• vast-runtime</li> </ul>                                                                                                                                                                                                                                                                                                                               |
| Pipeline  | The pipeline that emitted the log.                                                                                                                                                                                                                                                                                                                                                                                                             |
| Message   | A description of the logged event.                                                                                                                                                                                                                                                                                                                                                                                                             |

You can also inspect a log: Select a log in the Logs page and click the Inspect button to open the Inspect panel. The following details are shown:

- The log message, if it is long.
- The resource.
- A Trace ID, with a hyperlink to the expanded trace display.
- A span ID associated with the log.



- A mini version of the trace.

## Viewing Pipeline Traces

Traces are collected using the OpenTelemetry framework and observatory system, which enables you to see how each trigger event traverses through the pipeline, how time is distributed between function deployments, how traces are associated with specific attributes and logs, which function deployment events are invoked by a given trigger event, and provides visibility into performance bottlenecks.

You can view a list of traces and you can explore each trace for more visibility.

### Viewing All Traces

To see all traces, open the Pipeline Management page from the left navigation menu and select Traces.

To see traces for a specific pipeline, right-click the pipeline and select Pipeline Traces. This opens the Traces page and filters the traces for the pipeline you selected.

The following properties are shown per trace:

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| Start Time | The start time of the trace.                                            |
| Pipeline   | The pipeline that emitted the trace.                                    |
| Status     | The status of the trace.                                                |
| Trace ID   | The trace ID. Click the trace ID to <a href="#">explore the trace</a> . |
| Duration   | The duration of the trace.                                              |

### Exploring a Trace

When you click a trace ID, the user interface shows you one or more spans of the trace in the pipeline's runtime, as color coded bars.

To see logs associated with the trace, click Logs.

To see logs associated with a component span, click the relevant bar.

The following attributes are also displayed for the trace:

- Trace Start Time
- Duration
- Functions
- Depth
- Total Span

# Upgrading DataEngine Runtime Version

In the event that you need to upgrade DataEngine runtime, follow this procedure:

1. Run the [DataEngine CLI](#), and make sure it is updated to the latest builder image:
  - a. Use [vastde builders inspect](#) to check current builder image URL:

| Plaintext                                                                                                                                                                                                                                                                                                                             | Copy |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre>user@user-machine:~\$ vastde builders inspect Current Builder Image Configuration  Builder Image URL: https://github.com/vast- data/dataengine-cli/releases/tag/v5.4.1-dev.29d4871e  To change the builder image, use: vastde builders set &lt;image-url&gt; To view tag details, use: vastde builders inspect &lt;tag&gt;</pre> |      |

- b. Run [vastde builders list](#) to list available builder images.
  - c. Run [vastde builders set](#) to update the CLI configuration with the latest builder image URL:

| Plaintext                                                                                                                                                                                                                                                                                                                                                                                                                  | Copy |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <pre> user@user-machine:~\$ vastde builders set v5.4.1- dev.c0b8b3d5 Using default repository, full image: vastdataorg/vast- builder:v5.4.1-dev.c0b8b3d5 Builder image URL updated successfully!  Previous: https://github.com/vast-data/dataengine- cli/releases/tag/v5.4.1-dev.c0b8b3d5 Current:  vastdataorg/vast-builder:v5.4.1-dev.29d4871e  This builder image will be used for building DataEngine functions </pre> |      |

2. Rebuild each function and push it to the container registry. You can do this with [vastde functions build](#):

| Plaintext                                                                                                                 | Copy |
|---------------------------------------------------------------------------------------------------------------------------|------|
| <pre> user@user-machine:~\$ vastde functions build my-function --target ./my-function-code --image-tag v2.0 --push </pre> |      |

3. Update each function resource to point to its new image. You can do this from the DataEngine GUI:
  - a. From the left navigation menu, select **Manage Elements** (  ) and then **Functions**.
  - b. Right-click the function and select **Edit**.  
The revision number is incremented automatically from the previous revision.
  - c. In the **Image Tag** field, enter the image tag attached to the new version of the container image.
  - d. Modify other fields as needed:

|                             |                                                                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Revision Alias</b>       | An alias for the revision.                                                                                                                             |
| <b>Revision Description</b> | A description specifically for the revision.                                                                                                           |
| <b>Container Registry</b>   | The container registry where the image of the function is stored. If different from selected, select the correct container registry from the dropdown. |
| <b>Artifact Source</b>      | The path on the container registry where the image resides.                                                                                            |
| <b>Full image path</b>      | This is the full path to the image. It is formed automatically from the container registry, artifact source and image tag values that you supply.      |

e. Click **Update Function**.

A new revision is created and listed in the **Functions** page as a revision of the original function.

4. Redeploy each pipeline with new revisions of its functions:

- From the left navigation menu, select **Manage Pipelines** (  ) and then **Pipelines**.
- Right-click the pipeline and select **Go to Builder**.
- For each function:
  - Select the function in the builder.
  - Click **Settings** to show the function deployment configuration in the **Settings** pane on the right.
  - From the **Revision Number** dropdown, select the new revision of the function.
- When you are done updating function revisions and making any other changes, click **Deploy**.

The pipeline is redeployed.

5. Once all pipelines are redeployed with rebuilt function images, navigate to the dashboard (click ).

6. Verify that under the heading **System-**

**Wide Overview: Functions And Pipelines Performance And Trends**, you no longer see the error and you now see data. The graphs now include **Active Pipelines**, **Failing Pipelines** and **Slowest Pipelines**.